

Extraction of benzene from heptane with pyridinium based ionic liquid at (298.15, 308.15 and 318.15) K



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ABSTRACT

In this research, the liquid–liquid equilibrium data (LLE) for the ternary systems of {N-butylpyridinium nitrate, [BPy][NO₃] + heptane + benzene} and {N-hexylpyridinium nitrate, [HPy][NO₃] + heptane + benzene} at the temperatures (298.15, 308.15 and 318.15) K and atmospheric pressure were determined. The solute distribution ratio and selectivity from the LLE data were calculated. The benzene selectivity for systems with [BPy][NO₃] is higher than [HPy][NO₃] and [BPy][NO₃] is an appropriate solvent for benzene extraction. Besides, the influence of the temperature on benzene distribution ratio and selectivity of benzene was also investigated. Finally, the experimental data were correlated with the NRTL thermodynamic model.

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1. Introduction

Extraction of benzene from aliphatic hydrocarbon mixtures is one of the major importance processes in the petrochemical industry. These compounds have boiling points in a close range which leads to the formation of multiple azeotropic mixtures. Liquid extraction may be a suitable process for extraction of aromatic from aliphatic hydrocarbon. There are some polar organic solvents used for the extraction of aromatics from aliphatic hydrocarbons such as sulfolane, N-formyl morpholine (NFM) and N-methylpyrrolidone (NMP) [1–6]. Due to the toxicity, flammability and volatility of these solvents, they produce so many problems in the process.

In recent years, ionic liquids (ILs) were proposed as an environmentally friendly alternative to replace traditional organic solvents for liquid–liquid extraction processes [7]. The unique properties of ionic liquids include low vapor pressure, non-flammability, chemical and thermal stability, large liquid range, easy to recycle and high solubility for different types of compounds [8]. In the last decade, some researches on the extraction of benzene from aliphatic hydrocarbons using ILs have been published [9–24]. Most of the works, were done using imidazolium based ionic liquids, but a few studies were published with pyridinium

based ILs as compared to the imidazolium for extraction of benzene from alkanes [7,22–25]. The aim of this research is to evaluate the potential of pyridinium based ionic liquids for extraction of benzene from heptane.

This paper is a continuation of our ongoing research on the extraction of aromatic from their mixtures with alkane using ILs as solvents [26–28]. In this work, the liquid–liquid equilibria (LLE) data have been experimentally determined for the ternary systems of {N-butylpyridinium nitrate [BPy][NO₃] or N-hexylpyridinium nitrate [HPy][NO₃] + heptane + benzene} at temperatures range (298.15–318.15) K and atmospheric pressure. From these data, the benzene distribution ratio (β) and selectivity (S) were calculated and the suitability of the studied ILs was evaluated. Besides, the effect of the temperature on the efficiency of the separation process was also investigated. In addition, the LLE data was correlated with the NRTL activity coefficient model [29].

2. Experiments

2.1. Chemicals

The suppliers, water content, molecular weight and mass fraction purity of the compounds are reported in Table 1.

[BPy][NO₃] and [HPy][NO₃] were produced and purified in our laboratory. [BPy][NO₃] was prepared from its corresponding [BPy][Br] that was synthesized following the method described in the

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Table 1

The suppliers, water content, molecular weight and purities of the chemicals.

Chemical name	Supplier	Mass fraction purity	Water content (ppm)	M _w
Heptane	Merck	>0.99	121	100.2
Benzene	Fluka	>0.99	316	78.11
[BPy][NO ₃]	Synthesized	>0.98	1011	198.21
[HPy][NO ₃]	in our lab	>0.98	874	226.1

literature [30]. Alkylation was carried out in an inert atmosphere and $T = 338$ K by use of pyridine and a little excess of 1-Bromobutane without addition of solvent [30]. The desired N-butylpyridinium Nitrate was obtained from a metathetic reaction of N-butylpyridinium Bromide with silver nitrate [31]. The procedure method for synthesis of [HPy][NO₃] was the same as for [BPy][NO₃] with a modification 1-Bromohexane was used instead of 1-Bromobutane in alkylation stage. The structure of ILs was tested with nuclear magnetic resonance (NMR) spectroscopy (Figs. S1 and S2 in supplementary data). Before use, the ILs were inserted under vacuum for 24 h at moderate temperature (343 K) in order to be dry and degas. The purified ILs were kept in under argon gas to avoid moisture and their water content was measured with a Karl Fischer titration 684 model.

2.2. Apparatus and procedure

The experimental data for {[BPy][NO₃]} or [HPy]

Table 2Experimental data based on mole fraction, calculated benzene distribution ratio (β) and selectivity (S) values for ternary systems of {[BPy][NO₃]} (1) + heptane (2) + benzene (3) at temperatures (298.15, 308.15 and 318.15) K and $P = 89$ kPa.^a

Feed			Heptane rich phase		IL rich phase		β	S
x_1	x_2	x_3	x_2^I	x_3^I	x_2^{II}	x_3^{II}		
$T = 298.15$								
0.301	0.664	0.035	0.965	0.035	0.010	0.037	1.06	101.45
0.300	0.629	0.071	0.923	0.077	0.009	0.059	0.76	74.48
0.301	0.559	0.140	0.838	0.162	0.008	0.097	0.60	66.51
0.300	0.490	0.210	0.744	0.256	0.006	0.123	0.48	58.85
0.300	0.418	0.282	0.644	0.356	0.005	0.147	0.41	48.78
0.300	0.349	0.351	0.548	0.452	0.005	0.176	0.39	40.75
0.301	0.280	0.419	0.450	0.550	0.005	0.208	0.38	33.00
0.300	0.211	0.489	0.344	0.656	0.005	0.231	0.35	24.77
0.300	0.140	0.560	0.237	0.763	0.004	0.276	0.36	19.70
0.300	0.070	0.630	0.125	0.875	0.004	0.341	0.39	12.54
$T = 308.15$								
0.300	0.664	0.036	0.963	0.037	0.009	0.032	0.87	90.28
0.301	0.628	0.071	0.919	0.081	0.009	0.052	0.64	66.14
0.300	0.559	0.141	0.830	0.170	0.007	0.079	0.47	57.44
0.301	0.489	0.210	0.736	0.264	0.006	0.105	0.40	51.32
0.301	0.419	0.280	0.641	0.359	0.006	0.132	0.37	40.97
0.300	0.350	0.350	0.540	0.460	0.005	0.150	0.33	33.36
0.300	0.280	0.420	0.436	0.564	0.005	0.165	0.29	26.00
0.300	0.210	0.490	0.333	0.667	0.005	0.192	0.29	21.19
0.301	0.139	0.560	0.226	0.774	0.004	0.229	0.30	16.62
0.301	0.068	0.631	0.114	0.886	0.004	0.271	0.31	8.78
$T = 318.15$								
0.301	0.663	0.036	0.961	0.039	0.009	0.028	0.73	78.13
0.301	0.627	0.072	0.917	0.083	0.009	0.048	0.58	59.54
0.301	0.557	0.142	0.824	0.176	0.006	0.071	0.41	52.46
0.300	0.489	0.211	0.729	0.271	0.005	0.092	0.34	47.21
0.301	0.419	0.280	0.630	0.370	0.005	0.105	0.28	35.58
0.300	0.350	0.350	0.533	0.467	0.005	0.127	0.27	29.31
0.301	0.280	0.419	0.433	0.567	0.005	0.153	0.27	23.90
0.301	0.207	0.492	0.325	0.675	0.005	0.175	0.26	17.51
0.301	0.139	0.560	0.221	0.779	0.004	0.202	0.26	13.11
0.301	0.071	0.628	0.116	0.884	0.004	0.241	0.27	8.56

^a Standard uncertainties (u) are: $u(x_i) = \pm 0.001$, $u(T) = \pm 0.1$ K, $u(P) = \pm 1$ kPa.

[NO₃] + heptane + benzene} systems were measured at temperatures 298.15, 308.15 and 318.15 K. An appropriate amount of the heptane, benzene and ILs were weighted using a laboratory balance (OHAUS, Adventurer AV264C) and mixed into the glass vials with a volume of approximately 15 cm³. The experimental uncertainty for mass measurement was $\pm 10^{-4}$ g. The glass vials were hermetically closed with a cap and septum to prevent any losses of contents. The mixtures formed two liquid phases and inserted into a shaker (IKA HS-260) for 5 h at 300 rpm. The vials were placed in a water bath (Julabo, FP50) for 12 h in order to attain the thermodynamic equilibrium and complete phase separation. The precision of water bath was ± 0.1 K. After that, the compositions of the both phases were analyzed using a long needle syringe. The gas chromatography (Varian, cp 3800) was used for determination of phase compositions. The detector for gas chromatography was FID detector with capillary column (Chrompack, 30 m $\times$ 0.25 mm $\times$ 1.2 μ m).

Table 3Experimental data based on mole fraction, calculated benzene distribution ratio (β) and selectivity (S) values for ternary systems of {[HPy][NO₃]} (1) + heptane (2) + benzene (3) at the temperatures (298.15, 308.15 and 318.15) K and $P = 89$ kPa.^a

Feed			Heptane rich phase		IL rich phase		β	S
x_1	x_2	x_3	x_2^I	x_3^I	x_2^{II}	x_3^{II}		
$T = 298.15$								
0.301	0.664	0.035	0.968	0.032	0.027	0.042	1.28	45.28
0.300	0.630	0.070	0.930	0.070	0.024	0.072	1.03	40.88
0.300	0.558	0.142	0.852	0.148	0.021	0.129	0.87	36.29
0.301	0.488	0.211	0.759	0.241	0.018	0.160	0.67	27.96
0.301	0.419	0.280	0.671	0.329	0.017	0.201	0.61	24.50
0.300	0.350	0.350	0.578	0.422	0.016	0.244	0.58	20.59
0.301	0.279	0.421	0.480	0.520	0.017	0.292	0.56	16.34
0.301	0.209	0.490	0.376	0.624	0.014	0.333	0.53	14.24
0.301	0.141	0.559	0.268	0.732	0.013	0.384	0.52	11.08
0.300	0.070	0.630	0.147	0.853	0.010	0.455	0.53	8.10
$T = 308.15$								
0.301	0.663	0.036	0.966	0.034	0.028	0.040	1.16	40.37
0.301	0.628	0.071	0.926	0.074	0.023	0.064	0.87	35.41
0.300	0.559	0.141	0.841	0.159	0.019	0.104	0.65	29.47
0.300	0.488	0.212	0.748	0.252	0.018	0.139	0.55	22.79
0.300	0.420	0.280	0.658	0.342	0.019	0.176	0.51	18.04
0.300	0.350	0.350	0.560	0.440	0.017	0.208	0.47	15.17
0.300	0.280	0.420	0.462	0.538	0.016	0.249	0.46	13.48
0.301	0.210	0.489	0.363	0.637	0.014	0.301	0.47	11.97
0.301	0.140	0.559	0.252	0.748	0.013	0.346	0.46	8.95
0.301	0.071	0.628	0.135	0.865	0.011	0.410	0.47	5.64
$T = 318.15$								
0.301	0.663	0.036	0.962	0.038	0.024	0.032	0.82	32.81
0.301	0.628	0.071	0.920	0.080	0.020	0.052	0.65	30.21
0.301	0.558	0.141	0.833	0.167	0.017	0.091	0.55	27.11
0.300	0.488	0.212	0.740	0.260	0.016	0.120	0.46	20.71
0.301	0.420	0.279	0.646	0.354	0.016	0.146	0.41	16.50
0.301	0.350	0.349	0.547	0.453	0.016	0.172	0.38	12.77
0.301	0.279	0.420	0.450	0.550	0.015	0.217	0.40	11.67
0.300	0.210	0.490	0.346	0.654	0.013	0.251	0.38	9.96
0.301	0.141	0.559	0.241	0.759	0.014	0.306	0.40	7.15
0.300	0.070	0.630	0.125	0.875	0.011	0.365	0.42	4.69

^a Standard uncertainties (u) are: $u(x_i) = \pm 0.001$, $u(T) = \pm 0.1$ K, $u(P) = \pm 1$ kPa.

The compositions of the upper phase were heptane and benzene and no IL detected in this phase. This was confirmed by the NMR analysis. Determinations the top phase composition require the calibrations curve for the systems heptane and benzene. The calibration curve was prepared with injection of different mixtures of heptane and benzene into the GC. The bottom phase contains the IL, heptane and benzene. A sample of this phase was mixed with acetone in order to dilute the samples and injected into the GC. As the negligible vapor pressure of ILs, GC cannot detect the ILs and a

glass wool was put into the injector liner to collect the ILs. A sample of bottom phase was diluted with acetone (1:5) and then 0.5 μl of this mixture was injected into the GC where, glass wool was put into the injector liner. ILs is adsorbed by the glass wool and after 10–20 injection the glass wool was replaced with new one. It should be emphasized that the results of injection for different mixtures of heptane and benzene into the GC without glass wool were the same as injection with glass wool.

The standard uncertainties in mole fraction for the top and

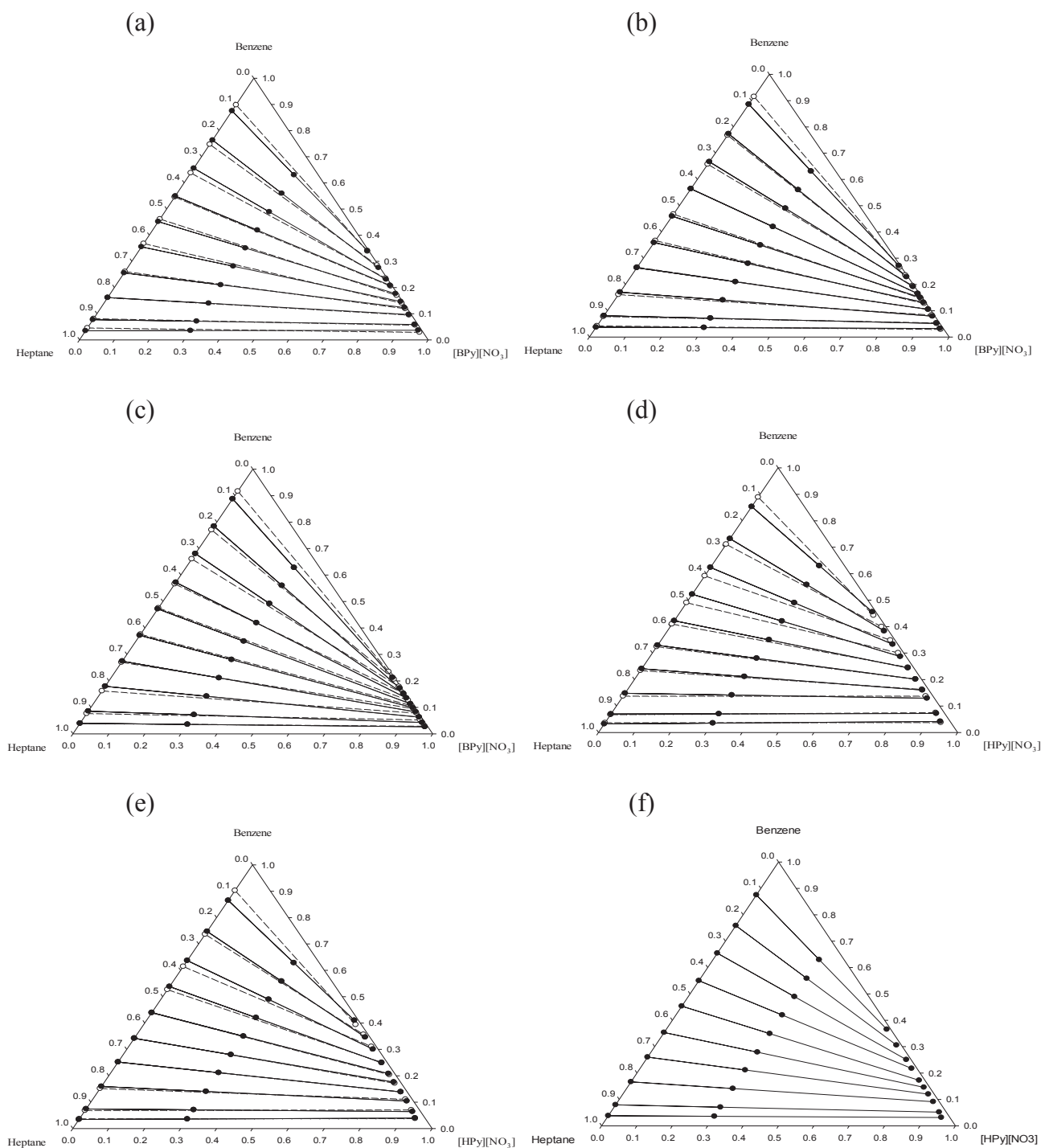


Fig. 1. Experimental LLE of the ternary systems ([BPyl][NO₃] (1) + heptane (2) + benzene (3)); (a) T = 298.15 K; (b) T = 308.15 K; (c) T = 318.15 K, and ([HPyl][NO₃] (1) + heptane (2) + benzene (3)); (d) T = 298.15 K; (e) T = 308.15 K; (f) T = 318.15 K. The solid lines show the experimental data and dashed line show the NRTL model.

bottom phase were less than $\pm 1 \times 10^{-3}$.

3. Result and discussion

3.1. Experimental data

Experimental results for the system of {[BPy][NO₃] or [HPy][NO₃] (1) + heptane (2) + benzene (3)} at (298.15, 308.15 and 318.15) K are reported in Tables 2 and 3. The triangular diagrams for studied systems are schemed in Fig. 1. As the figures show, the solubility of benzene is more than heptane in the tested ILs, which indicates [BPy][NO₃] and [HPy][NO₃] are potential solvents for the extraction of benzene from heptane. Comparing the ILs with the same [NO₃[−]] anion in this work, it was found that the [BPy]⁺ cation shows bigger immiscible region than [HPy]⁺ cation. Moreover, the effect of temperature is small in Fig. 1. The Comparison of the studied systems show that the immiscible region increases as the temperature rises, which is mostly visible in the IL-rich phase.

The ILs capacities may be evaluated by calculation the selectivity (S) and benzene distribution ratio (β) according to the following equations:

$$S = \frac{x_3^{\text{II}} x_2^{\text{I}}}{x_3^{\text{I}} x_2^{\text{II}}}, \quad (1)$$

$$\beta = \frac{x_3^{\text{II}}}{x_3^{\text{I}}}, \quad (2)$$

Where, x is the mole fraction and the subscripts 2 and 3 denote heptane and benzene, respectively. The selectivity values more than one means the ILs is suitable for solvent extraction.

The high values of benzene distribution ratio and selectivity mean the lower solvent and stages requirement for a separation process and so minimum operating costs and smaller apparatus [32]. The temperature effect on benzene distribution ratio and selectivity of benzene is represented in Figs. 2 and 3 (Figs. S3 and S4 in supplementary data). In these figures the benzene selectivity and the distribution ratios against the benzene mole fraction in heptane rich phase is represented. The figures are exhibited the extraction of benzene from heptane at low temperature (T = 298.15 K) is more efficiently, that mean the selectivity and benzene distribution ratio are increase with reducing temperature. This phenomenon was observed by Garcia et al. [15] for extraction of benzene from hexane using other ILs. The selectivity of benzene for both types of studied ILs in this work and different types of the IL for the system {IL (1) + heptane (2) + benzene (3)} were compared in Fig. 4. The values of benzene selectivity for [BPy][NO₃] are greater than [BMpyr][NTf₂] [16], [EPy][EtSO₄] [23], and [Bmim][NO₃] [27], which could be used for replacement of polar solvents. The selectivity values for systems with different cation and the same anion of ILs ([BPy][NO₃] - [Bmim][NO₃]) have been compared in Fig. 4. According to this figure, the selectivity values for pyridinium were rather greater than the imidazolium based ILs. Besides the selectivity of benzene for [BPy][NO₃] are higher than [HPy][NO₃]. This behavior was also observed for the selectivity of benzene for imidazolium based ILs ([Bmim][NO₃] and [Omim][NO₃]) [27], which means increasing the alkyl chain length of cation of ILs reduce the values of benzene selectivity. Also the comparison of the values of benzene distribution ratio for both types of studied ILs with mole fraction of benzene in the heptane rich phase was represented in Fig. 5. The values of benzene distribution ratio for [HPy][NO₃] are higher than the [BPy][NO₃], that imply, in equal conditions more amount of [BPy][NO₃] will be used as a solvent. This is not a serious problem because the ILs can be recovered and reused many times.

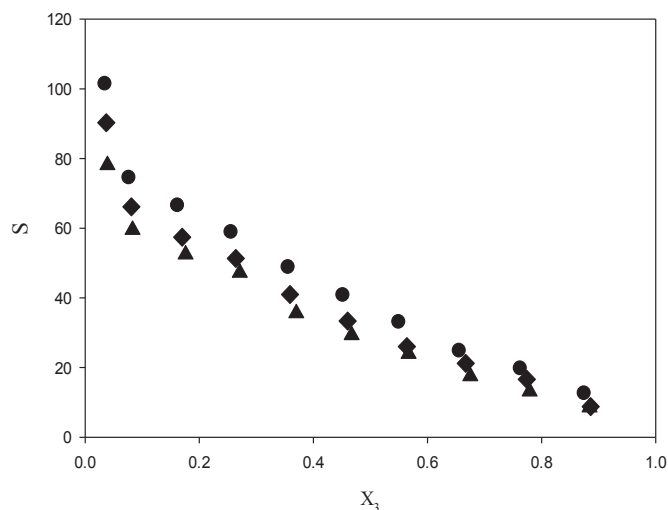


Fig. 2. Selectivity, as a function of the benzene mole fraction in the hydrocarbon-rich phase, for the ternary systems {[BPy][NO₃] + heptane + benzene}: (●) T = 298.15 K, (◆) T = 308.15 K, (▲) T = 318.15 K.

3.2. Thermodynamic correlation

The NRTL thermodynamic model [29] was applied for correlation the experimental data in this research. The activity coefficient for the component i of the system is expressed as follows:

$$\ln \gamma_i = \frac{\sum_{j=1}^n \tau_{ji} x_j G_{ji}}{\sum_{k=1}^n x_k G_{ki}} + \sum_{j=1}^n \left(\frac{x_j G_{ij}}{\sum_{k=1}^n x_k G_{kj}} \right) \left(\tau_{ij} - \frac{\sum_{m=1}^n \tau_{mi} x_m G_{mi}}{\sum_{k=1}^n x_k G_{kj}} \right), \quad (3)$$

where

$$G_{ij} = \exp(-\alpha \tau_{ij}), \quad (4)$$

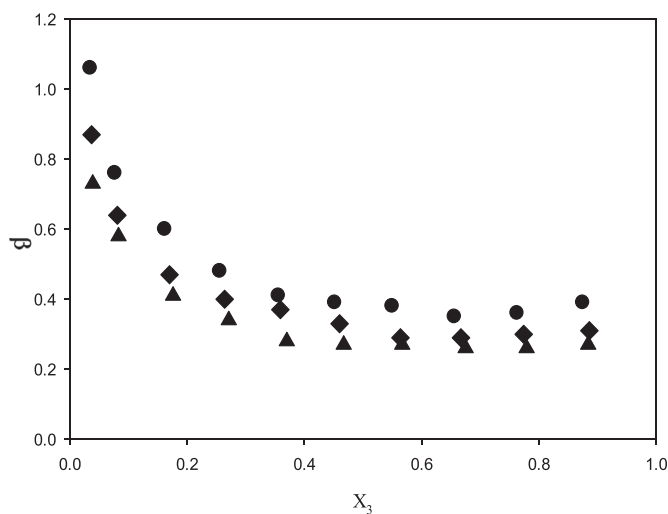


Fig. 3. Benzene distribution ratio as a function of the benzene mole fraction in the hydrocarbon-rich phase, for the ternary systems {[BPy][NO₃] + heptane + benzene}: (●) T = 298.15 K, (◆) T = 308.15 K, (▲) T = 318.15 K.

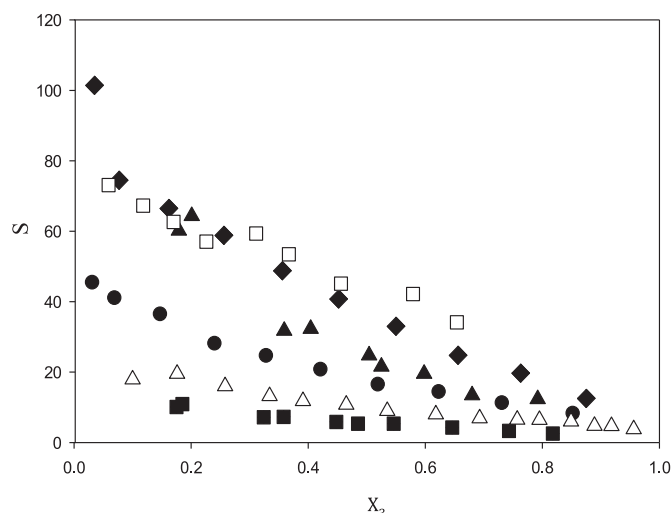


Fig. 4. The comparison of the selectivity of the experimental and literature data with mole fraction for the ternary systems (solvent (1) + heptane (2) + benzene (3)): ●, [HPy][NO₃] (this work); ◆, [BPy][NO₃] (this work); ■, [Omim][NO₃] [27]; ▲, [Bmim][NO₃] [27]; △, [BMpyr][NTf₂] [16]; □, [EPy][EtSO₄] [23].

$$\tau_{ij} = \frac{\Delta g_{ij}}{RT}, \quad (5)$$

Where, x is the mole fraction, R is the gas constant, T is absolute temperature, α is a non-randomness parameter which was set to 0.3, and Δg_{ij} is an energy parameter which is defined as:

$$\Delta g_{ij} = g_{ij} - g_{ji}, \quad (6)$$

A minimization between the experimental and calculated mole fraction differences of each component in both phases was carried out on an objective function which is defined as:

$$F = \sum_i^m \sum_j^{n-1} \sum_k^2 \left(\frac{x_{ijk}^{\text{exp}} - x_{ijk}^{\text{cal}}}{x_{ijk}^{\text{exp}}} \right)^2, \quad (7)$$

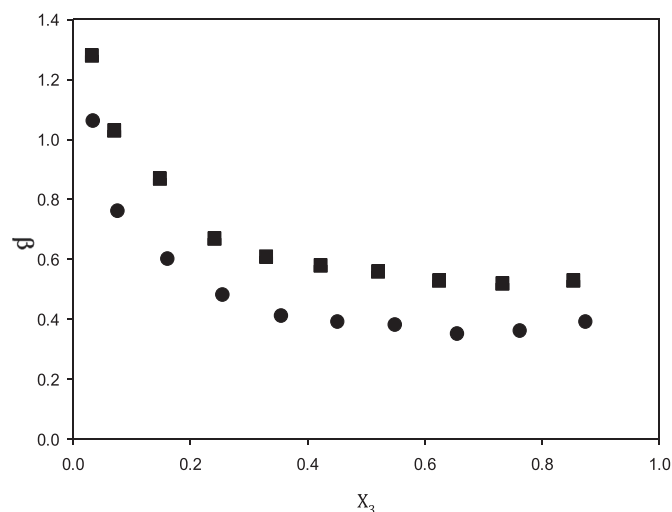


Fig. 5. The variation of the benzene distribution ratio versus benzene mole fraction in the heptane rich phase for the ternary system {[BPy][NO₃] or [HPy][NO₃] (1) + heptane (2) + benzene (3)} at $T = 298.15$ K, ● [BPy][NO₃], ■ [HPy][NO₃].

Table 4

The interaction parameters (Δg) for NRTL model and deviation based on *rmsd* for ternary systems of (IL + heptane + benzene).

Component i–j	NRTL parameter		rmsd
	Δg_{ij} /kJ. mol ⁻¹	Δg_{ji} /kJ. mol ⁻¹	
<i>[BPy][NO₃] (1) + heptane (2) + benzene (3)</i>			
1–2	19.240	–3.139	0.80
1–3	6.479	19.824	
2–3	5.931	20.692	
<i>[HPy][NO₃] (1) + heptane (2) + benzene (3)</i>			
1–2	13.467	–1.341	1.21
1–3	7.468	19.292	
2–3	7.298	17.485	

Where the number of tie lines, the number of components and the number of phases represents with m , n and k , respectively. The superscripts of *exp* and *cal* denote the experimental or the calculated mole fraction values. The root mean square deviation (*rmsd*) is expressed as follow:

$$rmsd = \left[\frac{\sum_i^m \sum_j^{n-1} \sum_k^2 \left(\frac{x_{ijk}^{\text{exp}} - x_{ijk}^{\text{cal}}}{x_{ijk}^{\text{exp}}} \right)^2}{2mn} \right]^{1/2} \times 100, \quad (8)$$

The estimated energy parameters and the *rmsd* are listed in Table 4. Based on the *rmsd* values reported in Table 4, the NRTL model fit the experimental data with good accuracy.

4. Conclusion

In this paper, new experimental data were obtained for the ternary mixtures of {[BPy][NO₃] (1) + heptane (2) + benzene (3)} and {[HPy][NO₃] (1) + heptane (2) + benzene (3)} at (298.15, 308.15 and 318.15) K and atmospheric pressure. The benzene selectivity and distribution ratio were calculated from experimental data. All systems showed selectivity values higher than the unity, which means these ILs could be used as solvents for extraction of benzene from heptane. The benzene selectivity in the [BPy][NO₃] systems are higher than [HPy][NO₃] which imply [BPy][NO₃] is suitable IL for benzene extraction. The temperature effect on the studied systems was also evaluated. The extraction of benzene from its mixture at low temperatures is more efficiently than high temperatures, which the temperature effect on efficiency of the separation process was found to be small. The LLE data was satisfactorily correlated with the NRTL model.

List of symbols

F	Objective function
K	Number of phases
M	Number of the tie lines
N	Number of the components
S	Selectivity
T	Temperature
w_w	water content by mass
x_i	mole fraction of component i

Greek letters

α	Non-randomness parameter in the NRTL model
β	Solute distribution ratio
Δg_{ij}	Energy parameter

Subscripts and superscripts

1, 2, i, j Components 1, 2, i, and j

I Alkane rich phase

II IL rich phase

cal. Calculated values

exp. Experimental values

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.fluid.2015.12.009>.

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